

# 4

## Homology groups

In this chapter we shall formalize the ideas sketched in the Introduction and define homology groups for an arbitrary oriented simplicial complex  $K$ . There is one group,  $H_p(K)$ , in each dimension  $p$  with  $0 \leq p \leq \dim K$ ; the group  $H_p(K)$  measures, roughly speaking, the number of ‘independent  $p$ -dimensional holes’ in  $K$ . If  $K$  is an oriented graph then  $H_1(K)$  is isomorphic to the group  $Z_1(K)$  of 1-cycles on  $K$  (see 1.19).

In order to read this chapter it is necessary to know about quotient groups and exact sequences, and to know a little about presentations. All these are explained in the appendix to the book.

### Chain groups and boundary homomorphisms

The chain groups associated with an oriented simplicial complex are exactly analogous to those associated with an oriented graph (1.17), and are defined as follows.

**Definition 4.1.** Let  $K$  be an oriented simplicial complex of dimension  $n$ , and let  $\alpha_p$  be the number of  $p$ -simplexes of  $K$ . For  $0 \leq p \leq n$ , let  $\sigma_p^1, \dots, \sigma_p^{\alpha_p}$  be the oriented  $p$ -simplexes of  $K$ ; for such  $p$  the  $p$ th chain group of  $K$ , denoted  $C_p(K)$ , is the free abelian group on the set  $\{\sigma_p^1, \dots, \sigma_p^{\alpha_p}\}$  (A.11). Thus an element of  $C_p(K)$  is a linear combination

$$\lambda_1 \sigma_p^1 + \dots + \lambda_{\alpha_p} \sigma_p^{\alpha_p}$$

with the  $\lambda$ ’s integers. This is called a  $p$ -chain on  $K$  and two  $p$ -chains are added by adding corresponding coefficients. When writing down  $p$ -chains it is customary to omit  $p$ -simplexes whose coefficient is zero – unless they are all zero, when

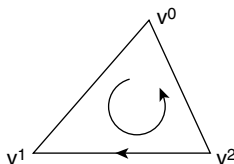
we just write 0. We also write  $\sigma$  for  $1\sigma$ . For  $p > n$  or  $p < 0$  the group  $C_p(K)$  is defined to be 0. (Notice that  $C_p(\emptyset) = 0$  for all  $p$ , since  $\emptyset$  has no simplexes at all.)

We shall shortly introduce a ‘boundary homomorphism’ which generalizes that for graphs (1.19). Recall that the boundary  $\partial(v^0v^1)$  of an oriented 1-simplex  $(v^0v^1)$  is the 0-chain  $v^1 - v^0$ . Now consider an oriented 2-simplex  $(v^0v^1v^2)$ . It is certainly desirable that the boundary should be the 1-chain

$$\partial(v^0v^1v^2) = (v^1v^2) + (v^2v^0) + (v^0v^1).$$

However, if the 2-simplex belongs to an oriented simplicial complex  $K$  then the three edges of the triangle will also have specific orientations as elements of  $K$ , and it might be that, for example, the bottom edge has orientation  $(v^2v^1)$ . In that case  $(v^1v^2) \notin K$  so that the above formula does not in fact give a 1-chain: we should really write

$$\partial(v^0v^1v^2) = -(v^2v^1) + \dots$$



Alternatively – and this is the course we adopt – we can make the convention that

$(v^1v^2)$  means the 1-chain  $(v^1v^2)$  if  $(v^1v^2) \in K$

and  $(v^1v^2)$  means the 1-chain  $-(v^2v^1)$  if  $(v^2v^1) \in K$ .

Then the above formula for  $\partial(v^0v^1v^2)$  always gives a 1-chain. In addition, this convention enables us to preserve the natural order of indices and write

$$\partial(v^0v^1v^2) = (v^1v^2) - (v^0v^2) + (v^0v^1).$$

More generally, the convention is as follows.

**Convention 4.2.** Let  $(v^0 \dots v^p)$  be an oriented  $p$ -simplex ( $p \geq 1$ ). Let  $K$  be an oriented simplicial complex and suppose that the oriented simplex  $\sigma = -(v^0 \dots v^p)$ , i.e.  $\sigma = (v^{\pi(0)} \dots v^{\pi(p)})$  for an odd permutation  $\pi$ , belongs to

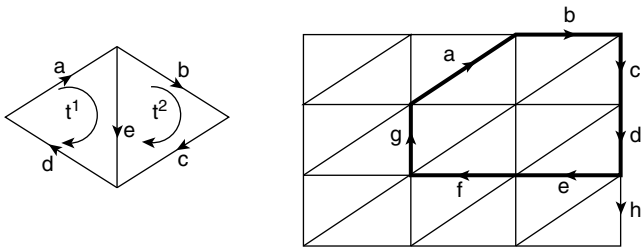
$$(v^0 \dots v^p) = -(v^{\pi(0)} \dots v^{\pi(p)})$$

**Definition 4.3.** Let  $\sigma = (v^0 \dots v^p)$  be an oriented  $p$ -simplex of  $K$  ( $K$  as in 4.1) for some  $p > 0$ . The *boundary* of  $\sigma$  is the  $(p - 1)$ -chain

where the hat over  $v^i$  indicates that it is omitted. Note that we are using the convention 4.2 here, for the face of  $\sigma$  with  $v^i$  omitted may or may not have the right orientation to be a  $(p - 1)$ -simplex of  $K$ . For  $p = 0$ ,  $\partial_p \sigma$  is defined to be zero. The *boundary homomorphism*  $\partial = \partial_p : C_p(K) \rightarrow C_{p-1}(K)$  is defined by

$$\partial_p(\Sigma \lambda_i \sigma_p^i) = \Sigma \lambda_i \partial_p(\sigma_p^i) \quad (\text{compare A.12})$$

As an example, consider the two coherently oriented triangles in the diagram on the left.



We have  $\partial(t^1 + t^2) = (a + e + d) + (b + c - e) = a + b + c + d$ , corresponding to the geometrical idea that the boundary of the whole area is the polygon  $a + b + c + d$ . Notice that  $\partial(a + b + c + d) = 0$ . Similarly, orienting all the triangles in the right-hand diagram clockwise, the boundary of the sum of the seven triangles enclosed by the heavy line is  $a + b + c + d + e + f + g$ , which is a 1-chain with boundary 0. If we make the usual identifications to turn the

simplicial complex on the right into a torus (see 2.2(3)) then  $\partial(c + d + h) = 0$ , but it can be shown quite easily that there is no 2-chain with boundary  $c + d + h$ : the simple closed polygon  $c + d + h$  goes round a hole in the torus and does not bound any region. Thus, on the torus,  $a + \dots + g$  bounds a region (is ‘homologous to zero’ as we shall say shortly) whereas  $c + d + h$  does not. The homology groups we study pick out the simple closed polygons – more generally the 1-cycles – which do not bound any 2-chain by taking the group of 1-cycles and factoring out by the subgroup consisting of those which do bound 2-chains.

**Remark 4.4.** It is far from obvious that the above definition of  $\partial\sigma$  makes sense – i.e. that it depends only on the oriented simplex  $\sigma$  and not on the particular ordering of vertices given. For instance,  $(v^0v^1v^2) = (v^1v^2v^0)$  (as oriented simplexes), and

$$\partial(v^0v^1v^2) = (v^1v^2) - (v^0v^2) + (v^0v^1);$$

$$\partial(v^1v^2v^0) = (v^2v^0) - (v^1v^0) + (v^1v^2).$$

These expressions are indeed equal (bearing 4.2 in mind), but perhaps some general reassurance is necessary. (The reassurance which follows could be omitted on a first reading.) We shall also check that the formula is compatible with the notation (3.12)  $-\sigma$  for the same simplex as  $\sigma$  but with opposite orientation. Both these things are contained in the statement

$$\partial(v^{\pi(0)} \dots v^{\pi(n)}) = \pm \partial(v^0 \dots v^n)$$

where the sign is  $+$  or  $-$  according as  $\pi$  is even or odd, and it is enough to verify this when  $\pi$  is a transposition (which will be odd).

Suppose, then, that  $n \geq 1$  and  $\pi$  transposes  $i$  and  $j$  where  $i < j$ . Consider the face of  $\sigma$  not containing the vertex  $v^j$ . This occurs in the formula for  $\partial(v^0 \dots v^n)$  as

$$(-1)^j(v^0 \dots v^i \dots \widehat{v^j} \dots v^n) = (-1)^j\tau$$

say and in the formula for  $\partial(v^{\pi(0)} \dots v^{\pi(n)})$  as

$$(-1)^i(v^0 \dots \widehat{v^j}, \dots v^i \dots v^n) = (-1)^i\tau' \quad \text{say.}$$

Now moving  $v^i$  to the left  $j - i - 1$  places turns  $\tau'$  into  $\tau$ . This amounts to effecting  $j - i - 1$  transpositions, so

$$(-1)^i\tau' = (-1)^i(-1)^{j-i-1}\tau = -(-1)^j\tau.$$

Similarly the face of  $\sigma$  not containing  $v^i$  occurs in the two formulae with opposite sign, while a face of  $\sigma$  containing both  $v^i$  and  $v^j$  occurs with the same power of  $-1$  but with  $v^i$  and  $v^j$  interchanged – hence altogether with opposite sign. This completes the verification.

**Definition 4.5.** The *augmentation*

$$\varepsilon : C_0(K) \rightarrow \mathbb{Z}$$

is the homomorphism defined by

$$\varepsilon(\sum \lambda_i \sigma_0^i) = \sum \lambda_i$$

(compare 1.22).

The sequences (not in general exact)

$$\dots 0 \rightarrow 0 \rightarrow C_n(K) \xrightarrow{\partial} C_{n-1}(K) \rightarrow \dots \rightarrow C_1(K) \xrightarrow{\partial} C_0(K) \rightarrow 0 \rightarrow 0 \dots$$

and

$$\dots 0 \rightarrow 0 \rightarrow C_n(K) \xrightarrow{\partial} C_{n-1}(K) \rightarrow \dots \rightarrow C_1(K) \xrightarrow{\partial} C_0(K) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0 \dots$$

are called the *chain complex* of  $K$  and the *augmented chain complex* of  $K$ , respectively. The main property of the boundary homomorphism is the following.

**Proposition 4.6.** For any  $p$ , the homomorphism

$$\partial \circ \partial = \partial_{p-1} \circ \partial_p : C_p(K) \rightarrow C_{p-2}(K)$$

is trivial. Also  $\varepsilon \circ \partial_1 : C_1(K) \rightarrow \mathbb{Z}$  is trivial.

**Proof** The last statement is straightforward to check; for the first one it is enough to prove that, for any  $p$ -simplex  $\sigma = (v^0 \dots v^p)$  of  $K$ , with  $p \geq 2$ ,

$$\partial_{p-1}(\partial_p \sigma) = 0.$$

Now

$$\partial_{p-1}(\partial_p \sigma) = \sum_{i=0}^p (-1)^i \partial_{p-1}(v^0 \dots \widehat{v^i} \dots v^p).$$

This is a sum of  $p(p+1)$  terms in which each  $(p-2)$ -dimensional face of  $\sigma$  occurs exactly twice. In fact, consider the  $(p-2)$ -dimensional face  $(v^0 \dots \widehat{v^i} \dots \widehat{v^j} \dots v^p)$ , where  $i < j$ . This occurs once in

$$(-1)^i \partial_{p-1}(v^0 \dots \widehat{v^i} \dots v^p), \text{ with sign } (-1)^i (-1)^{j-1}$$

and once in

$$(-1)^i \partial_{p-1}(v^0 \dots \widehat{v^j} \dots v^p), \text{ with sign } (-1)^j (-1)^i.$$

The total coefficient of this  $(p-2)$ -simplex in  $\partial_{p-1}(\partial_p \sigma)$  is therefore  $(-1)^i (-1)^{j-1} + (-1)^j (-1)^i = 0$ .  $\square$

**Corollary 4.7.** *For any  $p$ ,*

$$\text{Im } \partial_{p+1} \subset \text{Ker } \partial_p.$$

Also

$$\text{Im } \partial_1 \subset \text{Ker } \varepsilon.$$

$\square$

## Homology groups

**Definitions 4.8.** Consider the sequence

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K).$$

$\text{Ker } \partial_p$  is denoted by  $Z_p$ , or  $Z_p(K)$ , and elements of  $Z_p$  are called  $p$ -cycles (compare 1.19).  $Z_p$  is a free finitely generated (f.g.) abelian group. (See A.33)

$\text{Im } \partial_{p+1}$  is denoted by  $B_p$ , or  $B_p(K)$ , and elements of  $B_p$  are called  $p$ -boundaries (we also say that these elements *bound*).  $B_p$  is a free f.g. abelian group.

Thus the first statement of 4.7 can be summed up by the motto: ‘every boundary is a cycle’. The converse, ‘every cycle bounds’ is false in general – see the discussion preceding 4.4.

Since  $B_p \subset Z_p$  there is a quotient group  $H_p(K) = Z_p(K)/B_p(K)$ , called the  $p$ th homology group of  $K$ . It measures the extent to which  $K$  has non-bounding  $p$ -cycles i.e. the extent to which  $K$  has ‘ $p$ -dimensional holes’ – compare the Introduction, where some examples with  $p = 1$  were given. Note that  $H_p(K)$  is a finitely generated abelian group.

The quotient group  $\text{Ker } \varepsilon/B_0(K)$  is denoted by  $\tilde{H}_0(K)$  and is called the reduced zeroth homology group of  $K$ .

**4.9. Further terminology and remarks** (1) Let  $K^r$  denote the  $r$ -skeleton of  $K$  – that is the subcomplex of  $K$  consisting of all simplexes of dimension  $\leq r$ . Then clearly  $C_p(K) = C_p(K^r)$  and  $Z_p(K) = Z_p(K^r)$  provided  $r \geq p$ , and  $H_p(K) = H_p(K^r)$  provided  $r \geq p + 1$ .

(2) Taking  $r = 1$  in (1) we have  $Z_1(K) = Z_1(K^1)$ . Since  $K^1$  is an oriented graph we can use the method of Chapter 1 to write down a basis for  $Z_1(K^1)$ . Explicitly, when  $K$  is connected, a basis for  $Z_1(K) = Z_1(K^1)$  is obtained as follows (compare 1.20): choose a maximal tree  $T$  in  $K^1$  and let  $e^1, \dots, e^\mu$  be the edges of  $K$  not in  $T$ . Let  $z^i$  ( $i = 1, \dots, \mu$ ) be the 1-cycle associated to the unique loop on  $T + e^i$ , the coefficient of  $e^i$  in  $z^i$  being +1. Then  $z^1, \dots, z^\mu$  is the required basis. If  $K$  is not connected, then a basis is given by finding a basis for the cycles in each component.

Notice that, according to 1.23,  $\tilde{H}_0(K) = 0$  when  $K$  is connected.

(3) If  $K$  is  $n$ -dimensional, then  $C_{n+1}(K) = 0$  so that  $B_n(K) = 0$  and  $H_n(K) = Z_n(K)/0 \cong Z_n(K)$ . Perhaps it is being too pedantic to distinguish between  $Z_n$  and  $Z_n/0$ , so let us write  $H_n(K) = Z_n(K)$ . Note that  $Z_n$  is free abelian since it is a subgroup of  $C_n$  (see A.33): thus *top dimensional homology groups are free abelian*.

In the case  $n = 1$ , i.e.  $K$  is an oriented graph, we have  $H_1(K) = Z_1(K)$  so that, if  $K$  is connected,  $H_1(K) \cong \mathbb{Z}^{\mu(K)}$  and free generators of  $H_1(K)$  are given by the basic cycles described in (2) above. These are the ‘independent loops’ or ‘independent holes’ in the graph.

For any  $K$  of dimension  $n$ , we have  $H_p(K) = 0$  for  $p < 0$  and for  $p > n$ .

(4) An element of  $H_p(K)$  is a coset  $\bar{z} = z + B_p(K)$  where  $z \in Z_p(K)$ . We shall use the notation  $\{z\}$  for this coset; it is called the *homology class* of the cycle  $z$ . (If several simplicial complexes are under discussion at the same time, the notation  $\{z\}_K$  will be used.) Any cycle  $z'$  in  $\{z\}$  – that is, any cycle  $z' = z + b$  for  $b \in B_p(K)$  – is called a *representative cycle* for  $\{z\}$ . We say that  $z$  and  $z'$  are *homologous* if  $z' - z \in B_p(K)$ , and write  $z \sim z'$ ; in particular if  $z \in B_p(K)$  (i.e. if  $z$  bounds) we write  $z \sim 0$  :  $z$  is homologous to zero. Thus  $z \sim z' \Leftrightarrow \{z\} = \{z'\}$ . An element of  $C_p(K)$  which is not a cycle does not represent any homology class. Nevertheless we could (but won't) consider two chains to be homologous if their difference bounds. Note that a chain which bounds (is homologous to zero) is necessarily a cycle, by 4.7.

(5) As in the case of graphs (see 1.21(3)) the particular orientation given to a simplicial complex does not affect the homology groups up to isomorphism. The orientation is introduced in order to define the boundary homomorphism; nevertheless a theory does exist for unoriented simplicial complexes and this

theory is explained in Chapter 8. For simplicial complexes of dimension  $> 1$  the unoriented theory is a weaker tool than the oriented theory.

**Zeroth homology groups** Let  $K$  be an oriented simplicial complex; the object is to calculate  $H_0(K)$ . A vertex  $v$  of  $K$  can also be regarded as a 0-cycle (with the coefficient of  $v$  equal to 1 and other coefficients zero), and there is a simple geometrical interpretation of homology between vertices:

**Proposition 4.10.**  $v \sim v' \iff v$  and  $v'$  lie in the same component of  $K$ . (See 3.17(4))

**Proof** If  $v$  and  $v'$  lie in the same component of  $K$ , then a path from  $v'$  to  $v$  will have an associated 1-chain (1.18)  $c$  with the property  $\partial c = v - v'$ ; consequently  $v \sim v'$ . For the converse, we have to use  $v \sim v'$  to construct a path from  $v'$  to  $v$ . The reader can probably convince himself that this is possible by drawing a few pictures: alternatively here is a sketch of a proof by induction. Suppose that  $v - v' = \partial c$ , and let us write, temporarily,  $|c|$  for the sum of the moduli of the coefficients in  $c$ ; the induction is on  $|c|$  and the case  $|c| = 1$  is left to the reader. Suppose, to simplify notation, that all 1-simplexes of  $K$  with  $v$  as an end-point are oriented towards  $v$ . Then  $c$  must contain at least one such 1-simplex,  $e$  say, with positive coefficient. Writing  $e = (v''v)$ , we have  $|c - e| < |c|$  and  $\partial(c - e) = v'' - v'$  so that by induction there is a path from  $v'$  to  $v''$ , and hence via  $e$  to  $v$ .  $\square$

Now let  $K$  be a non-empty oriented simplicial complex with say  $k$  components  $K_1, \dots, K_k$ , and let  $v^i$  be a vertex in  $K_i$  for  $i = 1, \dots, k$ .

**Proposition 4.11.**  $H_0(K)$  is freely generated by the homology classes  $\{v^i\}$  for  $i = 1, \dots, k$ . Thus  $H_0(K) \cong \mathbb{Z}^k$  (also  $H_0(\emptyset) = 0$  since  $C_p(\emptyset) = 0$  for all  $p$ ).

**Proof** The implication ' $\Leftarrow$ ' of 4.10 shows that  $\{v^1\}, \dots, \{v^k\}$  generate  $H_0(K) = C_0(K)/B_0(K)$  (note that  $Z_0(K) = C_0(K)$ ). To show that they are free generators suppose  $\lambda_1 v^1 + \dots + \lambda_k v^k = \partial c$  for some  $c \in C_1(K)$ . Write  $c = c^1 + \dots + c^k$  where  $c^i$  contains only 1-simplexes in  $K_i$  ( $i = 1, \dots, k$ ). Then  $\partial c^i$  contains only vertices in  $K_i$ , so that we deduce  $\lambda_i v^i = \partial c^i$  for  $i = 1, \dots, k$ . Taking the augmentation (4.5) of both sides, and using (4.6)  $\varepsilon \circ \partial = 0$  we find  $\lambda_i = 0$  for  $i = 1, \dots, k$ .  $\square$

This shows that  $H_0(K)$  measures a very fundamental geometrical property of  $K$ , namely the number of separate pieces (components) it has.

**Remark 4.12.** In fact it is always true that, in the above notation,  $H_p(K) \cong H_p(K_1) \oplus \dots \oplus H_p(K_k)$  for all  $p$ . This is a special case of a theorem (7.2) to be



proved later on homology groups of unions of not necessarily disjoint simplicial complexes. It is also not hard to give a direct proof (start with  $k = 2$ ).

Now consider the sequence

$$C_1(K) \xrightarrow{\partial} C_0(K) \xrightarrow{\varepsilon} \mathbb{Z}.$$

We know that  $\text{Im } \partial \subset \text{Ker } \varepsilon \subset C_0(K)$  by 4.7, and applying A.16, Ex(7), this gives a short exact sequence

$$0 \rightarrow \frac{\text{Ker } \varepsilon}{\text{Im } \partial} \rightarrow \frac{C_0}{\text{Im } \partial} \rightarrow \frac{C_0}{\text{Ker } \varepsilon} \rightarrow 0$$

i.e.

$$0 \rightarrow \tilde{H}_0(K) \rightarrow H_0(K) \rightarrow \frac{C_0}{\text{Ker } \varepsilon} \rightarrow 0.$$

Now suppose that  $K$  is non-empty; then  $\varepsilon$  is epic (surjective) and  $C_0/\text{Ker } \varepsilon \cong \text{Im } \varepsilon = \mathbb{Z}$ . Hence the sequence is split (A.19), and the following result holds.

**Proposition 4.13.** *For a non-empty  $K$ ,*

$$H_0(K) \cong \tilde{H}_0(K) \oplus \mathbb{Z}.$$

Also

$$H_0(\emptyset) = \tilde{H}_0(\emptyset) = 0.$$

□

Note that this confirms that, for a connected  $K$ ,  $\tilde{H}_0(K) = 0$  (4.9(2)). Indeed the main reason for introducing  $\tilde{H}_0(K)$  is that it is often 0 and this makes calculations with exact sequences more straightforward. In the notation of 4.11, the classes represented by  $v^i - v^1$  for  $i = 2, \dots, k$  freely generate  $\tilde{H}_0(K)$ , when  $k \geq 2$ .

**First homology groups** Given an oriented connected simplicial complex  $K$  we now seek to calculate  $H_1(K)$ . When  $K$  is a graph ( $\dim K = 1$ ) there is no problem: we just calculate  $Z_1(K)$  as in Chapter 1 (see 4.9(2) and (3)). Thus in this case  $H_1(K) \cong \mathbb{Z}^{\mu(K)}$  with generators  $\{z^1\}, \dots, \{z^\mu\}$ .

In general there is a nontrivial subgroup  $B_1(K)$  of  $Z_1(K)$  to factor out:

$$C_2(K) \xrightarrow{\partial_2} C_1(K) \xrightarrow{\partial_1} C_0(K)$$

$$H_1(K) = \text{Ker } \partial_1 / \text{Im } \partial_2 = Z_1(K) / B_1(K).$$

To calculate this quotient in a particular instance we use the method of presentations; see the Appendix but note that the examples we actually meet are very

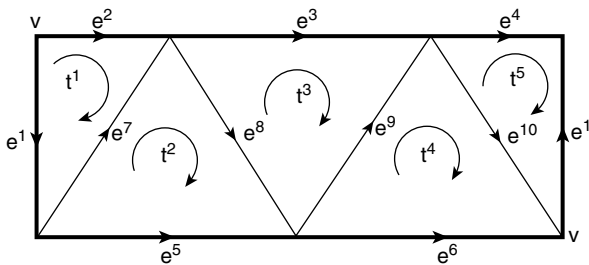
simple and require only a minimum of technique. The example in A.25 is a typical calculation. First we find a basis for  $Z_1(K) = Z_1(K^1)$  by the method of 4.9(2), then we find generators for  $B_1(K)$ . The generators we use are  $\partial_2 t$  where  $t$  runs over the oriented 2-simplexes of  $K$ . A presentation of  $H_1(K)$  is, when  $K$  is connected,

4.14

(cycles  $z^1, \dots, z^\mu, \mu = \mu(K^1)$ , associated to  
basic loops on  $K^1$  (see 4.9(2))  
a relation  $\partial_2 t = 0$  for each oriented 2-simplex of  $K$ .

Direct calculations using 4.14 are rather tedious, and we shall not need to make many of them. Here is an example.

**4.15. Möbius Band** The diagram shows a triangulation  $K$  of the Möbius band, and the heavily drawn edges are ones whose removal leaves a maximal tree in  $K^1$ . It is a good idea to label these edges  $e^1, \dots, e^\mu$  (in this example  $\mu = 6$ ) and the remaining edges  $e^{\mu+1}, \dots, e^\alpha (\alpha = \alpha_1(K))$ .



The reason for this is as follows. Suppose that  $z$  is any 1-cycle on  $K$ ; then by 1.20

4.16

$$\begin{aligned} z &= \lambda_1 e^1 + \dots + \lambda_\mu e^\mu + \lambda_{\mu+1} e^{\mu+1} + \dots + \lambda_\alpha e^\alpha \\ \Rightarrow z &= \lambda_1 z^1 + \dots + \lambda_\mu z^\mu \end{aligned}$$

where  $z^i$  ( $i = 1, \dots, \mu$ ) is the 1-cycle associated with the unique loop on (maximal tree  $\cup e^i$ ), the 1-cycle being oriented so that it contains  $e^i$  with coefficient  $+1$ . This is of great assistance when expressing  $\partial_2 t$  as a linear combination of  $z^1, \dots, z^\mu$ .

Let  $z^1, \dots, z^6$  be as above. Thus

$$z^1 = e^1 + e^7 + e^8 + e^9 + e^{10}, \quad z^2 = e^2 + e^8 + e^9 + e^{10}$$

and so on, but there is no need to write them all out. In fact,  $B_1(K)$  is generated by the following elements of  $Z_1(K)$

$$\partial_2 t^1 = e^2 - e^7 - e^1 = z^2 - z^1 \quad (\text{see 4.16 above})$$

$$\partial_2 t^2 = e^7 + e^8 - e^5 = -z^5$$

$$\partial_2 t^3 = e^3 - e^9 - e^8 = z^3$$

$$\partial_2 t^4 = e^9 + e^{10} - e^6 = -z^6$$

$$\partial_2 t^5 = e^4 - e^1 - e^{10} = z^4 - z^1.$$

Thus  $H_1(K)$  has presentation

$$\left( z^1, z^2, z^3, z^4, z^5, z^6 \right. \\ \left. z^2 - z^1 = -z^5 = z^3 = -z^6 = z^4 - z^1 = 0. \right.$$

(Remember that the coset symbol  $\{ \}$  is conventionally omitted when writing down presentations; see A.21.) This shows that all the generators can be expressed in terms of  $z^1$ , and (compare A.24),  $H_1(K)$  is isomorphic to the abelian group with presentation

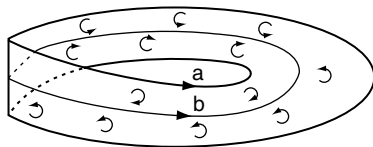
$$\left( z^1 \right. \\ \left. [\text{no relations}] \right.$$

which is infinite cyclic. Furthermore it shows that  $\{z^1\}_K$  generates  $H_1(K)$ . Using the relations alternative generators are  $\{z^2\}$ , or  $\{z^4\}$ , or  $\{z^2 + z^3\} \dots$

Notice that the cycle  $z^1$  goes ‘once round’ the Möbius band. Thus the result says that a 1-cycle going ‘once round’ does not bound anything, and that any 1-cycle is, up to homology, a multiple of this one. For example  $e^2 + e^3 + e^4 + e^5 + e^6$  is a 1-cycle going round the rim of the band, and

$$\begin{aligned} \{e^2 + e^3 + e^4 + e^5 + e^6\} &= \{z^2 + z^3 + z^4 + z^5 + z^6\} \quad \text{by 4.16} \\ &= \{z^2\} + \{z^3\} + \{z^4\} + \{z^5\} + \{z^6\} \\ &= \{z^1\} + 0 + \{z^1\} + 0 + 0 \\ &\quad \text{from the presentation of } H_1(K) \\ &= 2\{z^1\}. \end{aligned}$$

Indeed it is geometrically clear that the rim is a circle going twice round before joining up. The following picture shows more informally that the rim  $a$  is homologous to twice a circle  $b$  going round the middle of the band.



The orientations shown are coherent (2.3) except across the centre line  $b$ . Thus  $a - 2b$  bounds the whole oriented area (i.e. if the area were broken up into triangles, oriented as shown, the sum of their boundaries would be  $a - 2b$ ), so that  $a \sim 2b$ .

We know from 4.11 that  $H_0(K) \cong \mathbb{Z}$ , with generator the homology class of any vertex. To find  $H_2(K)$  we need  $\text{Ker}(\partial_2 : C_2(K) \rightarrow C_1(K))$ , since  $C_3(K) = 0$ . Now a general element of  $C_2(K)$  is  $c = \sum_{i=1}^5 \lambda_i t^i$ ; suppose that  $\partial_2 c = 0$ . The edge  $e^2$  occurs only in  $\partial_2 t^1$ ; hence  $t^1$  must have coefficient zero in  $c$ . Similar arguments show that  $t^2, t^3, t^4, t^5$  all have coefficient zero in  $c$ , so that  $c = 0$  :  $H_2(K) = 0$ . An alternative way of doing this is to write down  $\partial c$  explicitly, using  $\partial t^1 = e^2 - e^7 - e^1$  etc., and to deduce from  $\partial c = 0$  that  $\lambda_1 = \dots = \lambda_5 = 0$ .

Thus  $H_0(K) \cong \mathbb{Z}$  generator  $\{v\}$

$$H_1(K) \cong \mathbb{Z} \text{ generator } \{e^1 + e^7 + e^8 + e^9 + e^{10}\}$$

$$H_2(K) = 0.$$

**Examples 4.17.** (1X) Triangulate the cylinder (3.22(3)) and show that the homology groups are

$$H_0 \cong \mathbb{Z}; H_1 \cong \mathbb{Z} \text{ (generator a cycle going round one end of the cylinder)}$$

$$H_2 = 0.$$

Thus up to isomorphism the groups are the same as for the Möbius band. This fact becomes obvious by the method of collapsing (3.30) applied to homology groups (6.6). See 6.7(1).

(2X) Show that, for any  $p$ ,

$$C_p(K) \cong Z_p(K) \oplus B_{p-1}(K).$$

[Hint. Form a short exact sequence from  $\partial_p : C_p(K) \rightarrow C_{p-1}(K)$ .]

(3X) Using  $B_p(K) \subset Z_p(K) \subset C_p(K)$  and A.16, Ex. (7) show that

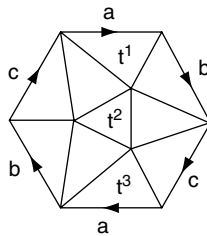
$$C_p(K)/B_p(K) \cong H_p(K) \oplus B_{p-1}(K).$$

This shows that the torsion subgroup of  $H_p(K)$  (the subgroup of  $H_p(K)$  consisting of those elements which are of finite order) is isomorphic to the torsion subgroup of  $C_p/B_p$  (since  $B_{p-1}$  is free abelian). In particular the torsion subgroup of  $H_p(K)$  is determined by the homomorphism  $\partial_{p+1} : C_{p+1}(K) \rightarrow C_p(K)$ .

Calculate the group  $C_1(K)/B_1(K)$  when  $K$  is an  $n$ -sided polygon. Note that the result depends on  $n$ , whereas  $H_1(K) \cong \mathbb{Z}$  for any  $n$  since  $\mu(K) = 1$  (see 4.9(3)). For example if  $K$  is an  $n$ -sided polygon then  $C_1(K)/B_1(K)$  is isomorphic to  $\mathbb{Z}^n$ , either because  $B_1(K)$  is trivial or, using the above isomorphism,  $H_1(K) \cong \mathbb{Z}$ ,  $B_0(K) \cong \mathbb{Z}^{n-1}$ , generated by the boundaries of  $n - 1$  of the edges.

(4X) Calculate the homology groups of the torus given in 2.2(3), or in 3.22(1). (Of course you must first orient the 1- and 2-simplexes, but it doesn't matter how – see 4.9(5).) This homology calculation is about as complicated as one normally wants to do 'from first principles'. Verify that  $H_0 \cong \mathbb{Z}$ ;  $H_1 \cong \mathbb{Z} \oplus \mathbb{Z}$  with generators  $\{a + b + c\}$  and  $\{d + e + f\}$  in 2.2(3);  $H_2 \cong \mathbb{Z}$  with generator the sum of all triangles of the torus, coherently oriented.

(5X) Orient the simplicial complex shown, which is a projective plane (2.7(3)), and verify that the homology groups are  $H_0 \cong \mathbb{Z}$ ;  $H_1 \cong \mathbb{Z}_2$  with generator  $\{a + b + c\}$ ;  $H_2 = 0$ .



Note that  $H_1$  is a finite group here; in fact  $2(a+b+c) = 0$ , i.e.  $2(a+b+c) \sim 0$ . This can be seen directly by adding up the boundaries of the ten triangles in the picture, all oriented clockwise. (The three labelled triangles are for later reference.)

We shall not have to calculate, directly from the definition, many homology groups  $H_p(K)$  for  $p > 1$ . There is one case of interest, however, in connexion with the orientability of closed surfaces.

**Proposition 4.18.** *Let  $K$  be a closed surface (2.1 or 3.29(2)). Then*

$$H_2(K) \cong \begin{cases} 0 & \text{if } K \text{ is non-orientable (2.3)} \\ \mathbb{Z} & \text{if } K \text{ is orientable.} \end{cases}$$

*A generator for  $H_2(K)$ , when  $K$  is orientable, is given by the homology class of the sum of all 2-simplexes of  $K$ , coherently oriented.*

To be precise:  $K$  is already oriented, and, apart from this, the 2-simplexes can be coherently oriented. The generator of  $H_2(K)$  is  $\Sigma(\pm t^i)$ , the sum being over all oriented 2-simplexes  $t^i$  of  $K$ , with sign  $+$  or  $-$  according as the orientation of  $t^i$  as a simplex of  $K$  does or does not agree with that in the chosen coherent orientation. Note that  $H_2(K) = Z_2(K)$  since  $K$  is two-dimensional (see 4.9(3)). A generator of  $H_2(K)$ , when  $K$  is an orientable closed surface, is called a *fundamental cycle*, or *fundamental class* for  $K$ . Up to sign it is determined by  $K$ .

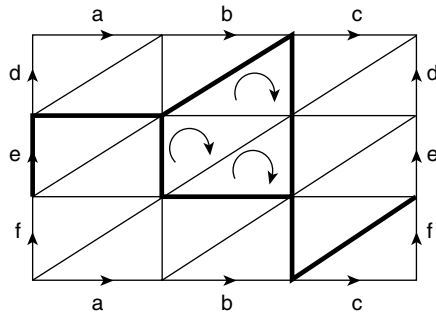
**Proof** Suppose first that  $K$  is orientable. For simplicity let us suppose that the orientation of  $K$  is such that the 2-simplexes receive orientations which are already coherent. We seek  $Z_2(K)$  (compare 4.9(3)), so let  $z \in Z_2(K)$ , and suppose  $z \neq 0$  so that some oriented 2-simplex  $t$  occurs with coefficient  $\lambda \neq 0$  in  $z$ . Let  $t'$  be any other oriented 2-simplex. Then (see 2.2(5)) there is a sequence of triangles of  $K$  connecting  $t$  and  $t'$ , two consecutive members of the sequence always having an edge in common. From this and  $\partial z = 0$  it follows that every triangle in the sequence, and in particular  $t'$ , also has coefficient  $\lambda$  in  $z$ . Hence  $z$  is a multiple of the sum of all triangles, coherently oriented, and  $Z_2(K)$  is infinite cyclic with this sum as generator.

Now suppose that  $H_2(K) \neq 0$  and let  $z$  be a non-zero element of  $Z_2(K)$ . Exactly as above we can deduce that the coefficient of each oriented 2-simplex in  $z$  must be, up to sign, the same non-zero integer (only up to sign because the triangles of  $K$  are no longer assumed coherently oriented). Now define a new orientation of the triangles of  $K$  by changing the orientation of those with negative coefficient in  $z$ . It is easy to see (using  $\partial z = 0$ ) that this coherently orients the triangles of  $K$ :  $K$  is orientable.  $\square$

Thus  $H_2$  distinguishes algebraically between orientable and non-orientable surfaces. Fundamental cycles for the tori drawn in 2.2(3) and 3.22(1) are obtained by orienting all the triangles in the pictures clockwise, or alternatively all anti-clockwise. Of course this doesn't work for the Klein bottle or projective plane of 2.2(4), for such an orientation will not be coherent across, for example, the edge  $d$ .

## Relative homology groups

In Chapter 9 we shall investigate some of the properties of a graph which is realized as a subcomplex of a surface. This is a situation involving a pair of simplicial complexes – or, as we shall say, a relative situation – and calls for algebraic tools which involve both complexes at once. In particular consider the graph in a torus drawn below.



There is a region bounded by five edges and containing the three oriented triangles. The sum of the boundaries of these three triangles is certainly not zero, but it does contain only edges in the graph, and the sum of the three triangles is called a relative cycle. Thus relative cycles have a connexion with regions, and this is exploited in Chapter 9.

**Definition 4.19.** Let  $K$  be an oriented simplicial complex of dimension  $n$  and  $L$  be a subcomplex of  $K$  ( $L$  is therefore oriented in a natural way by  $K$ ). For any  $p$ ,  $0 \leq p \leq n$ , the  $p$ th chain group of  $K$  modulo  $L$ , or the  $p$ th relative chain group (of the pair  $(K, L)$ ) is the subgroup of  $C_p(K)$  consisting of those  $p$ -chains on  $K$  in which the coefficient of every simplex of  $L$  is zero. It is denoted  $C_p(K, L)$ , and an element of  $C_p(K, L)$  is called a  $p$ -chain of  $K$  modulo  $L$ , or a relative  $p$ -chain. (There is always a choice of terminology between ‘relative’ and ‘modulo’. In the definitions below we shall usually give the former and leave the reader to give expression to the latter. Also ‘modulo’, when it occurs, will always be abbreviated to ‘mod’.) For  $p > n$  or  $p < 0$ ,  $C_p(K, L)$  is defined to be zero.

Note that  $C_p(K, \emptyset) = C_p(K)$ . The following definition of ‘relative boundary’ also reduces to the old definition of boundary when  $L = \emptyset$ .

**Definition 4.20.** With the notation of 4.19, we define

$$j = j_q : C_q(K) \rightarrow C_q(K, L)$$

to be the homomorphism, which changes to zero (or leaves at zero) the coefficient of every simplex in  $L$  (for  $q < 0$  or  $q > n$ ,  $j_q = 0$ ). Now define

$$\begin{aligned} \partial &= \widehat{\partial}_p : C_p(K, L) \rightarrow C_{p-1}(K, L) \\ \text{by } \widehat{\partial}c &= j_{p-1}(\partial_p c) \quad \text{for all } c \in C_p(K, L) \end{aligned}$$

$\widehat{\partial}$  called the *relative boundary homomorphism*. Thus if  $i_p$  denotes the inclusion of  $C_p(K, L)$  in  $C_p(K)$ , then  $\widehat{\partial}_p = j_{p-1} \circ \partial_p \circ i_p$ . We may say that  $j$  ‘forgets  $L$ ’

and then the relative boundary of a chain is the ordinary boundary followed by forgetting  $L$ .

**Proposition 4.21.** *For any  $p$ , the homomorphism*

$$\widehat{\partial} \circ \widehat{\partial} = \widehat{\partial}_{p-1} \circ \widehat{\partial}_p : C_p(K, L) \rightarrow C_{p-2}(K, L)$$

*is trivial.* □

The proof is similar to that of 4.6 and will be omitted.

The remaining definitions are precisely analogous to those for the ‘absolute’ case  $L = \emptyset$ . They are summarized below.

**Definitions 4.22.** *Chain complex of  $(K, L)$ :*

$$\begin{aligned} \dots 0 \rightarrow C_n(K, L) \xrightarrow{\widehat{\partial}} C_{n-1}(K, L) \xrightarrow{\widehat{\partial}} \dots \\ \xrightarrow{\widehat{\partial}} C_1(K, L) \xrightarrow{\widehat{\partial}} C_0(K, L) \rightarrow 0. \end{aligned}$$

(There is no augmented relative chain complex.)

Group of *relative  $p$ -cycles*  $= Z_p(K, L) = \text{Ker } \widehat{\partial}_p$ .

Group of *relative  $p$ -boundaries*  $= B_p(K, L) = \text{Im } \widehat{\partial}_{p+1}$ .

By 4.21,  $B_p(K, L) \subset Z_p(K, L)$ ; the quotient group

$$H_p(K, L) = Z_p(K, L) / B_p(K, L)$$

is the  $p$ th *relative homology group* of the pair  $(K, L)$ . All these reduce to the previous definitions when  $L = \emptyset$ .

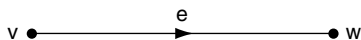
If  $z \in Z_p(K, L)$  then  $z + B_p(K, L) \in H_p(K, L)$  is denoted by  $\{z\}$  or  $\{z\}_{K, L}$  and is called the *relative homology class* of  $z$ . Any element of  $\{z\}$ , i.e. any  $z + b$  for  $b \in B_p(K, L)$ , is called a *representative relative cycle* for  $\{z\}$ . Two relative cycles  $z$  and  $z'$  are said to be *homologous mod  $L$*  (written  $z \stackrel{L}{\sim} z'$ ) if  $\{z\} = \{z'\}$ .

**Remarks and Examples 4.23.** (1) If  $K = L$  then all the groups and homomorphisms are trivial. If  $\dim K = n$ , then  $H_p(K, L) = 0$  for  $p < 0$  or  $p > n$ , and  $H_n(K, L) = Z_n(K, L)$  is free abelian.

(2)  $H_p(K, L)$  is not in general a subgroup of  $H_p(K)$ . For example let  $K$  be a 1-simplex  $e = (vw)$  together with its end-points and let  $L$  be the end-points. Then  $H_1(K) = 0$  since  $K$  is a tree (see the preamble to 4.14), but  $H_1(K, L) \cong \mathbb{Z}$ ,



with generator  $\{e\}$ . For  $C_0(K, L) = 0$ , so  $\widehat{\partial}_1 = 0$  and  $Z_1(K, L) = C_1(K, L)$  is infinite cyclic with generator  $e$ .



(3) More generally than (2), let  $K$  be any oriented graph and  $L = K^0$ , the set of vertices of  $K$ . Then  $C_0(K, L) = 0$ ,  $\widehat{\partial}_1 = 0$  and  $Z_1(K, L) = C_1(K, L)$  is a free abelian group with one generator for each edge of  $K$ .

If  $K$  is any oriented connected graph and  $L$  is a maximal tree (1.10) for  $K$ , then  $L$  contains all the vertices of  $K$  and so  $C_0(K, L) = 0$ ,  $\widehat{\partial}_1 = 0$  and  $Z_1(K, L) = C_1(K, L)$  is a free abelian group with one generator for each edge of  $K$  not in  $L$  – hence of rank  $\mu(K)$ .

(4) Let  $K^r$  denote the  $r$ -skeleton of  $K$ . Then the chain complex of  $(K^r, K^{r-1})$  is

$$\dots \rightarrow 0 \rightarrow C_r(K^r, K^{r-1}) \rightarrow 0 \rightarrow \dots$$

with only one non-zero group. Thus  $H_p(K^r, K^{r-1})$  is zero when  $p \neq r$ , and is free abelian of rank  $\alpha_r(K)$  (the number of  $r$ -simplexes of  $K$ ) when  $p = r$ . As an example, let  $K$  be a 2-simplex and all its faces, and  $r = 2$ , so that  $K^r = K$ ,  $K^{r-1}$  = the set of edges and vertices of  $K$ . Then  $H_2(K, K^1)$  is free abelian of rank 1, generated by the homology class of the 2-simplex. In this case  $H_1(K, K^1) = 0 = H_0(K, K^1)$ .

The next proposition enables relative cycles and boundaries to be recognized as such without undue difficulty. It is convenient, now and later, to make the following convention. When  $L$  is a subcomplex of  $K$ , we regard  $C_p(L)$  as a subgroup of  $C_p(K)$ , namely the subgroup consisting of those  $p$ -chains on  $K$  in which the coefficient of every  $p$ -simplex not in  $L$  is zero. (Thus, with this convention,  $C_p(K)$  is the internal direct sum of  $C_p(L)$  and  $C_p(K, L)$ .) In order that the convention shall not lead us into error, it is essential that the boundary of a  $p$ -chain on  $L$  should be the same whether we regard it in its own right or according to the convention as a special kind of  $p$ -chain on  $K$ . That this is indeed the case follows from the fact that  $L$  is a subcomplex of  $K$  and that therefore if  $\sigma \in L$  then all the faces of  $\sigma$  also belong to  $L$ .

**Proposition 4.24.** (1) Let, in the usual notation,  $c \in C_p(K, L)$ . Then  $c$  is a relative cycle, i.e. belongs to  $Z_p(K, L)$ , if and only if  $\partial c \in C_{p-1}(L)$ .

(2) Let  $c \in C_p(K, L)$ . Then  $c$  is a relative boundary, i.e.  $c \in B_p(K, L)$ , if and only if there is an (absolute)  $p$ -chain  $c' \in C_{p+1}(K)$  such that  $\partial c' - c \in C_p(L)$ .

**Proof** (1) is virtually immediate from the definitions (and the above convention), so the proof will be omitted. (2) can be a very confusing statement to verify, so here is a full proof.

Suppose that  $c \in B_p(K, L)$ , i.e.  $c = \widehat{\partial}c'$  for some  $c' \in C_{p+1}(K, L)$ . We show that this  $c'$  (which does of course belong to  $C_{p+1}(K)$ ) satisfies  $\partial c' - c \in C_p(L)$ . In fact

$$jc = c = \widehat{\partial}c' = j\partial c', \quad \text{so that} \quad j(c - \partial c') = 0.$$

From the definition of  $j$ , this implies  $\partial c' - c \in C_p(L)$ .

Suppose conversely that  $c'$ , as given in (2), exists. Then let  $c'' = jc' \in C_{p+1}(K, L)$ . We show  $\widehat{\partial}c'' = c$ , which proves that  $c \in B_p(K, L)$ .

We shall use the following simple but useful trick. Let  $\rho : C_{p+1}(K) \rightarrow C_{p+1}(L)$  denote the 'restriction' homomorphism which puts at zero the coefficient of any simplex not in  $L$ . Then clearly, for any  $c' \in C_{p+1}(K)$ ,  $c' = jc' + \rho c'$ .

We are given  $j(\partial c' - c) = 0$ ; also  $jc = c$  since  $c \in C_p(K, L)$ . Hence

$$\begin{aligned} c = jc = j\partial c' &= j\partial jc' + j\partial \rho c' = j\partial jc' \quad \text{since } \partial \rho c' \in C_p(L) \\ &= \widehat{\partial}c'' \quad \text{by definition.} \end{aligned}$$

□

**Zerorth relative homology groups** As an example of 4.24, let  $K$  be connected and  $L$  non-empty. We show  $H_0(K, L) = 0$ , i.e.  $B_0(K, L) = C_0(K, L)$ . Let  $v$  be a vertex of  $K$  not in  $L$  (if no such vertex exists then the result is true since  $C_0(K, L) = 0$ ), and  $w$  a vertex of  $L$ . Join  $w$  to  $v$  by a path in  $K$ ; the associated 1-chain (1.18)  $c'$  has  $\partial c' = v - w$ , so that  $\partial c' - v \in C_0(L)$ , i.e. (by 4.24)  $v \in B_0(K, L)$ . Hence every vertex of  $K$  not in  $L$  is a relative boundary, which implies the result.

The general result on  $H_0(K, L)$  is as follows. Details of the proof are left to the reader.

**Proposition 4.25.** *Let  $K$  be an oriented simplicial complex and  $L$  a subcomplex of  $K$ . Then  $H_0(K, L) = 0$  provided  $L$  has at least one vertex in every component of  $K$ , but  $H_0(K, L)$  acquires an infinite cyclic summand for every component of  $K$  not containing any vertex of  $L$ . Free generators for  $H_0(K, L)$  are the relative homology classes of one vertex from each of these latter components.* □

**Remark 4.26.** (compare 4.12) Let  $K$  have components  $K_1, \dots, K_k$  and write  $L_1 = L \cap K_1, \dots, L_k = L \cap K_k$ . Then it can be shown that, for all  $p$ ,

$$H_p(K, L) \cong H_p(K_1, L_1) \oplus \dots \oplus H_p(K_k, L_k).$$

**First relative homology groups** To calculate  $H_1(K, L)$  we need a basis for  $Z_1(K, L)$  and generators for  $B_1(K, L)$ . The method of presentations will then give the result as in the absolute case. The easiest case is when all the vertices of  $K$  belong to  $L$ , for then  $C_0(K, L) = 0$  so that  $Z_1(K, L) = C_1(K, L)$  and a basis is given by the edges of  $K$  not in  $L$ . Generators for  $B_1(K, L)$  are simply  $\widehat{\partial}_2 t$  where  $t$  runs through the oriented triangles of  $K$  not in  $L$ .

For completeness, here is a method for writing down a basis of  $Z_1(K, L)$  in general, followed by a very brief sketch of the proof. We assume  $K$  is connected (compare 4.26) to simplify the statement.

Let  $K$  be a connected oriented simplicial complex and  $L$  a subcomplex of  $K$ , with components  $L_1, \dots, L_\ell$ , say. Let  $T_\alpha$  be a maximal tree in the 1-skeleton of  $L_\alpha$  ( $\alpha = 1, \dots, \ell$ ) and let  $T$  be a maximal tree in the 1-skeleton of  $K$  which contains  $T_1, \dots, T_\ell$  (compare 1.12(1)). For each edge  $e^i$  of  $K$  which is in neither  $T$  nor  $L$  let  $z^i$  be the 1-cycle on  $K$  given by the unique loop on  $T + e^i$  such that the coefficient of  $e^i$  in  $z^i$  is +1. Thus the number of  $z^i$  is

$$m = \mu(K) - \mu(L_1) - \dots - \mu(L_\ell).$$

Next let  $v^1, \dots, v^\ell$  be vertices of  $K$  in, respectively, the components  $L_1, \dots, L_\ell$  of  $L$ , and let  $c^\alpha$  ( $\alpha = 2, \dots, \ell$ ) be the 1-chain associated to some path in  $K$  from  $v^1$  to  $v^\alpha$ . (Thus if  $\ell = 0$  or  $\ell = 1$  there is no  $c^\alpha$  to consider.)

**Proposition 4.27.** ★ *The following elements of  $Z_1(K, L)$  are free generators:*

$$jz^1, \dots, jz^m, jc^2, \dots, jc^\ell.$$

**Sketch of Proof** The proof depends on the following sequence.

$$0 \rightarrow Z_1(L) \xrightarrow{i} Z_1(K) \xrightarrow{j} Z_1(K, L) \xrightarrow{\partial_*} H_0(L) \xrightarrow{i_*} H_0(K) \rightarrow 0$$

where  $i$  = inclusion,  $j$  is the  $j$  of 4.20 restricted to  $Z_1(K)$ ,  $\partial_* z = \{\partial z\}_L$ ,  $i_* \{w\}_L = \{w\}_K$ . The sequence is a special case of the ‘homology sequence of a pair’ (see 6.3(6)) and is exact, provided  $L$  is non-empty. The case  $L = \emptyset$  ( $\ell = 0$ ) is covered by 1.20. The proof of exactness will be left to the reader – or he can wait for the general theorem. From the exact sequence we deduce two short exact sequences:

$$0 \rightarrow \text{Im } j \xrightarrow{i} Z_1(K, L) \xrightarrow{\partial_*} \text{Im } \partial_* \rightarrow 0 \quad (*)$$

$$\text{and } 0 \rightarrow Z_1(L) \xrightarrow{i} Z_1(K) \xrightarrow{j} \text{Im } j \rightarrow 0 \quad (**)$$

(each  $i$  being an inclusion).

Free generators for  $Z_1(L)$  are obtained by putting back, one by one, the edges of  $L_1$  not in  $T_1$ , then the edges of  $L_2$  not in  $T_2$ , etc. Free generators for  $Z_1(K)$  are all of these, plus  $z^1, \dots, z^m$ . It follows from  $(**)$  that free generators for  $\text{Im } j$  are  $jz^1, \dots, jz^m$ .

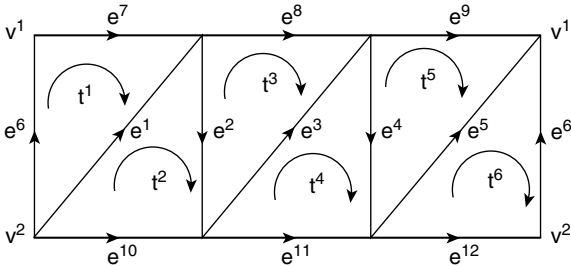
Free generators for  $\text{Im } \partial_* = \text{Ker } i_*$  are  $\{v^2 - v^1\}, \{v^3 - v^1\}, \dots, \{v^\ell - v^1\}$ , since  $i_*(\lambda_1\{v^1\}_L + \dots + \lambda_\ell\{v^\ell\}_L) = (\lambda_1 + \dots + \lambda_\ell)\{v^1\}_K$ . Finally,  $\partial_*jc^\alpha = \partial_*(c^\alpha - \rho c^\alpha)$  (where  $\rho$  is the ‘restriction’ which puts at zero the coefficients of simplexes not in  $L$ )  $= \{\partial c^\alpha\}_L - \{\partial \rho c^\alpha\}_L = \{v^\alpha - v^1\}_L - 0$ , since  $\rho c^\alpha \in C_1(L)$  so that  $\{\partial \rho c^\alpha\}_L = 0$ . Thus  $jc^2, \dots, jc^\ell$  are elements of  $Z_1(K, L)$  which go, under  $\partial_*$ , to the free generators of  $\text{Im } \partial_*$ . The result follows from  $(*)$  and the last sentence of A.20(3). ★

**Remark 4.28.** ★ The following method for expressing a given  $z \in Z_1(K, L)$  as an integer linear combination of the free generators listed in 4.27 is a straightforward consequence of the above proof.

Coefficient of  $jz^i =$  coefficient of  $e^i$  in  $z$  ( $i = 1, \dots, m$ ).

Coefficient of  $jc^\alpha =$  coefficient of  $\{v^\alpha\}$  in  $\partial_*z$  ( $\alpha = 2, \dots, \ell$ )  
= the sum of the coefficients of those vertices of  
 $L^\alpha$  in  $\partial z \in C_0(L)$ .★

**Examples 4.29.** (1) Let  $K$  be the cylinder drawn below and  $L$  the (heavily drawn) subcomplex consisting of the two ends.



In this case  $L$  contains all the vertices of  $K$ , so that  $C_0(K, L) = 0$ , and  $Z_1(K, L) = C_1(K, L)$  with basis  $e^1, \dots, e^6$ . Generators for  $B_1(K, L)$  are  $\widehat{\partial}t^1, \dots, \widehat{\partial}t^6$  and a

presentation for  $H_1(K, L)$  is

$$\left( \begin{array}{l} e^1, \dots, e^6 \\ -e^1 + e^6 = e^1 + e^2 = -e^2 - e^3 = e^3 + e^4 = -e^4 - e^5 = e^5 - e^6 = 0 \end{array} \right)$$

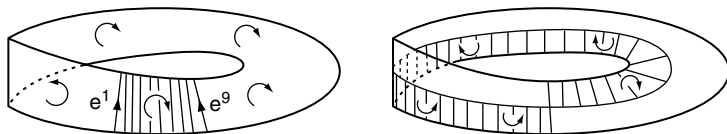
Thus  $H_1(K, L)$  is infinite cyclic, with generator  $\{e^6\}$  for example.

★ If we use 4.27 to obtain a basis for  $Z_1(K, L)$ , we have  $\ell = 2$  and taking  $L_1$  to be the top of the cylinder and  $L_2$  the bottom, we can take  $T_1 = \{e^7, e^8, v^1, v^3, v^4\}$ ,  $T_2 = \{e^{10}, e^{11}, v^2, v^5, v^6\}$  and  $T = T_1 \cup T_2 \cup \{e^6\}$ . Then  $m = 5$  and the cycles  $z^1, \dots, z^5$  from  $z$  are obtained from  $T$  by replacing, respectively say, the edges  $e^1, \dots, e^5$ . The first five generators of  $Z_1(K, L)$  are then  $e^1 - e^6, e^2 + e^6, e^3 - e^6, e^4 + e^6, e^5 - e^6$ . The remaining generator is, say,  $e^2$ , which is  $jc^2$  where  $c^2 = e^7 + e^2 - e^{10}$  is associated to a path from  $v^1 \in L_1$  to  $v^2 \in L_2$ . To express say  $\widehat{\partial}t^4 = e^3 + e^4$  in terms of these we use 4.28: the coefficients of  $jz^3$  and  $jz^4$  are both 1, and the coefficient of  $jc^2$  is 0 since  $\partial(e^3 + e^4) = v^6 - v^5$ , with sum of coefficients 0.

I leave it to the reader to verify that  $H_2(K, L) \cong \mathbb{Z}$  with generator  $\{t^1 + \dots + t^6\}$ . Of course  $H_0(K, L) = 0$  (4.25). ★

★(2X) Take  $K$  as in (1), but  $L$  this time to be one end, say the top, of the cylinder. Here  $L$  does not contain all the vertices of  $K$ , so some technique, e.g. that of 4.27, is really needed to write down generators for  $Z_1(K, L)$ . Verify that  $H_1(K, L)$  and  $H_2(K, L)$  are both trivial. ★

(3X) Take  $K$  to be the Möbius band given in 4.15 and  $L$  to be the rim of the band consisting of the edges  $e^2, \dots, e^6$  and their end-points. Verify that  $H_1(K, L) \cong \mathbb{Z}_2$ , with generator  $\{e^1\}$ . Notice that  $e^1 + e^1 \stackrel{L}{\sim} 0$ ; to see this geometrically refer to the left-hand diagram below.



Now  $e^1 \stackrel{L}{\sim} e^9$  since the boundary of the shaded area (oriented as shown) is, apart from pieces of  $L$ ,  $e^1 - e^9$ . But also  $e^1 \stackrel{L}{\sim} -e^9$  since the boundary of the rest of the band, oriented as shown, is  $e^1 + e^9$  apart from pieces of  $L$ . Hence  $e^1 \stackrel{L}{\sim} -e^1$ .

Verify that  $e^1 + e^7 + e^8 + e^9 + e^{10}$ , which goes ‘once round’ the band is homologous mod  $L$  to  $e^1$ . This can be seen geometrically by considering the boundary of the shaded region in the right-hand diagram.

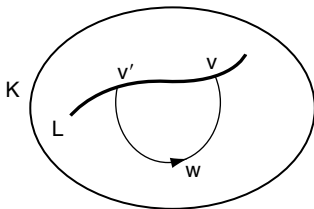
(4X) Consider the homomorphism  $j : C_1(K) \rightarrow C_1(K, L)$ . If  $z \in Z_1(K)$  then  $jz \in Z_1(K, L)$ , for

$$\begin{aligned}\widehat{\partial}jz &= j\partial jz = j\partial(z - \rho z) \quad (\rho : C_1(K) \rightarrow C_1(L) \text{ puts coefficients of} \\ &\quad \text{simplexes not in } L \text{ equal to } 0) \\ &= 0 - j\partial\rho z \text{ since } \partial z = 0 \\ &= 0 \text{ since } \partial\rho z = C_0(L).\end{aligned}$$

Thus there is an induced homomorphism, which we still call  $j$ ,

$$j : Z_1(K) \rightarrow Z_1(K, L)$$

(this appeared in the proof of 4.27). Suppose that  $L$  is connected (it does not matter whether  $K$  is connected). Show that this  $j$  is epic (surjective), i.e. that every relative 1-cycle can be ‘extended’, by adding edges of  $L$ , to an absolute 1-cycle. This is plausible from the picture, where the relative cycle  $w$  can be extended to an absolute cycle by adding the part of  $L$  between the endpoints  $v$  and  $v'$  of  $w$ .



[Hint for the formal proof: let  $\varepsilon_K : C_0(K) \rightarrow \mathbb{Z}$  and  $\varepsilon_L : C_0(L) \rightarrow \mathbb{Z}$  be the augmentations (4.5). Show that  $\varepsilon_L \partial w = \varepsilon_K \partial w = 0$  (see 4.6) and then use 1.23, which says that  $C_1(L) \xrightarrow{\partial} C_0(L) \xrightarrow{\varepsilon_L} \mathbb{Z}$  is exact.] The result is false when  $L$  is not connected. For example the relative cycle  $e^1$  on the cylinder in (1) above cannot be extended to an absolute cycle by adding edges in  $L$ .

(5X) Take  $K$  to be the torus given in 2.2(3), and  $L$  to be some particular subcomplex of  $K$ . Verify that  $H_2(K, L)$  is free of rank equal to the number of regions into which  $L$  divides  $K$  (in the obvious intuitive sense of ‘region’). Note that when  $L$  is the 1-skeleton of  $K$  this is a special case of 4.23(4), for the regions are then the 2-simplexes.

This will not work in general for a non-orientable surface such as the Klein bottle. For example take  $L = \emptyset$ ; then there is one region but  $H_2(K, L) = H_2(K) = 0$  by 4.18. More of this in Chapter 9.

### Three homomorphisms

We shall define three homomorphisms

$$i_* : H_p(L) \rightarrow H_p(K)$$

$$j_* : H_p(K) \rightarrow H_p(K, L)$$

$$\partial_* : H_p(K, L) \rightarrow H_{p-1}(L)$$

which, in the next chapter, will be fitted together to form an exact sequence. An alternative approach is sketched in an Appendix to this chapter. As usual  $K$  will be an oriented simplicial complex and  $L$  a subcomplex of  $K$ .

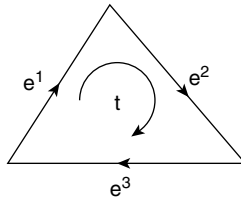
#### 4.30. The homomorphism $i_*$

Define  $i_* : H_p(L) \rightarrow H_p(K)$

$$\text{by } i_*\{z\}_L = \{z\}_K \quad (z \in Z_p(L))$$

$$\text{i.e. } i_*(z + B_p(L)) = z + B_p(K).$$

Thus we simply regard  $z \in Z_p(L)$  as a cycle on  $K$  and take its homology class there. Of course  $\{z\}_K$  can be zero when  $\{z\}_L$  is non-zero. For example let  $K$  be a 2-simplex and all its faces, and  $L$  the boundary of the 2-simplex (the set of proper faces). Then  $\{e_1 + e_2 + e_3\}_L$  generates  $H_1(K) \cong \mathbb{Z}$ , but  $\{e^1 + e^2 + e^3\}_K = 0$  since  $\partial t = e^1 + e^2 + e^3$ .



Several things must be checked before the definition of  $i_*$  is acceptable. These are

- (1) If  $z \in Z_p(L)$  then  $z \in Z_p(K)$  (recall the preamble to 4.24).
- (2) If  $z \in B_p(L)$  then  $z \in B_p(K)$ .
- (3)  $\{z + z'\}_K = \{z\}_K + \{z'\}_K$  where  $z, z' \in Z_p(L)$ .

(1) and (2) say that  $i_*$  is well-defined, and (3) say that  $i_*$  is a homomorphism.

All these assertions are straightforward to check.

**Examples 4.31.** (1) Let  $K$  be the Möbius band given in 4.15 and let  $L$  be the rim of the band. Then  $H_1(L) \cong \mathbb{Z}$ , with generator  $\{e^2 + e^3 + e^4 + e^5 + e^6\}_L$ , and  $H_1(K) \cong \mathbb{Z}$ , with generator  $\{z^1\}_K = \{e^1 + e^7 + e^8 + e^9 + e^{10}\}_K$  (see 4.15). The homomorphism  $i_*$  is determined by its effect on the generator of  $H_1(L)$ , and according to the calculation given in 4.15,

$$\begin{aligned} i_*\{e^2 + e^3 + e^4 + e^5 + e^6\}_L &= \{e^2 + e^3 + e^4 + e^5 + e^6\}_K \\ &= 2\{z^1\}_K. \end{aligned}$$

Thus  $i_*$  takes a generator to twice a generator: it is ‘multiplication by two’.

(2X) Let  $K$  be the cylinder given in 4.29(1), (2) and let  $L$  be (i) both ends; (ii) one end of the cylinder. Show that  $i_*$  takes (i) each of the generators  $\{e^7 + e^8 + e^9\}_L$  and  $\{e^{10} + e^{11} + e^{12}\}_L$  of  $H_1(L) \cong \mathbb{Z} \oplus \mathbb{Z}$  to a generator of  $H_1(K)$ ; (ii) the generator  $\{e^7 + e^8 + e^9\}_L$  of  $H_1(L) \cong \mathbb{Z}$  to a generator of  $H_1(K)$ . What is  $\text{Ker } i_*$  in case (i)?

(3X) Suppose that  $K$  has components  $K_1, \dots, K_k$  and that in addition  $L \cap K_1, \dots, L \cap K_k$  are connected (possibly empty). Show that  $i_* : H_0(L) \rightarrow H_0(K)$  is a monomorphism (injective). When is it an isomorphism? (In fact it is an isomorphism precisely when  $H_0(K, L) = 0$ ; in particular when  $K$  and  $L$  are both connected and non-empty.)

(4X) Let  $K$  be the oriented simplicial complex obtained from the projective plane drawn in 4.17(5) by removing the three 2-simplexes  $t^1, t^2, t^3$  (and then orienting the remaining simplexes). Let  $L$  be the subcomplex of  $K$  consisting of the eight 1-simplexes and five 0-simplexes on the boundaries of these three 2-simplexes. Find four 1-cycles  $z^1, z^2, z^3, z^4$  which freely generate  $Z_1(L)$ . Calculate  $H_1(K)$ , including generators, and express in terms of them the four homology classes  $\{z^\alpha\}_K, \alpha = 1, 2, 3, 4$ . Hence find  $i_* : H_1(L) \rightarrow H_1(K)$ ; in particular show that it is epic (surjective).

**4.32. The homomorphism  $j_*$**  Define  $j_* : H_p(K) \rightarrow H_p(K, L)$  by  $j_*\{z\}_K = \{jz\}_{K,L}$  for  $z \in Z_p(K)$ .

This time the following have to be checked:

- (1) If  $z \in Z_p(K)$  then  $jz \in Z_p(K, L)$  (see 4.20 for  $j$ ).
- (2) If  $z \in B_p(K)$  then  $jz \in B_p(K, L)$ .
- (3)  $\{j(z + z')\}_{K,L} = \{jz\}_{K,L} + \{jz'\}_{K,L}$ .

(The proof of (2) is very similar to that of 4.24(2).)



**Examples 4.33.** (1X) Take  $K$  and  $L$  as in 4.31(1). Then  $H_1(K, L) \cong \mathbb{Z}_2$  with generator  $\{e^1\}$  (see 4.29(3)). Thus  $j_*\{e^1 + e^7 + e^8 + e^9 + e^{10}\}_K = \{e^1\} + \{e^7\} + \{e^8\} + \{e^9\} + \{e^{10}\}$  (all on  $(K, L)$ )  $= 5\{e^1\} = \{e^1\}$ . Thus  $j_*$  takes generator to generator. What is  $j_*\{e^6 + e^2 + e^3 + e^4 + e^5\}_K$ ?

(2X) Take  $K$  and  $L$  to be as in (i) of 4.31(2). Show that  $j_* : H_1(K) \rightarrow H_1(K, L)$  is trivial.

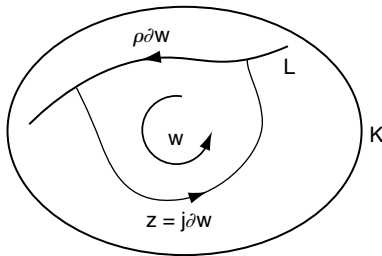
**4.34. The homomorphism  $\partial_*$**  Define  $\partial_* : H_p(K, L) \rightarrow H_{p-1}(L)$

$$\text{by } \partial_*\{z\}_{K,L} = \{\partial z\}_L \quad z \in Z_p(K, L).$$

This time the following have to be checked:

- (1) If  $z \in Z_p(K, L)$  then  $\partial z \in Z_{p-1}(L)$ .
- (2) If  $z \in B_p(K, L)$  then  $\partial z \in B_{p-1}(L)$ .
- (3)  $\{\partial z + \partial z'\}_L = \{\partial z\}_L + \{\partial z'\}_L$ .

From 4.24(1),  $\partial z \in C_{p-1}(L)$ ; however  $\partial \partial z = 0$  so  $\partial z \in Z_{p-1}(L)$ . Notice, incidentally, that although  $\partial z$  is a boundary,  $\{\partial z\}_L$  is not automatically zero. This is because  $\partial z$  is not usually the boundary of something in  $L$  – though of course it might be, as in the picture of 4.29(4), where  $z$  appears as  $w$ . For (2), suppose  $z = \widehat{\partial} w$  ( $w \in C_{p+1}(K, L)$ ); show that  $\partial z = \partial(-\rho \partial w)$ , where  $\rho : C_p(K) \rightarrow C_p(L)$  puts coefficients of simplexes not in  $L$  equal to 0. (See the diagram below.) The assertion (3) is immediate.



**Examples 4.35.** (1) Take  $K, L$  as in 4.31(1). Then  $\partial_* : H_1(K, L) \rightarrow H_0(L)$  is trivial – hardly surprising, for any homomorphism  $\mathbb{Z}_2 \rightarrow \mathbb{Z}$  is trivial.

(2X) Take  $K, L$  as in (i) of 4.31(2). Then  $\partial_* : H_2(K, L) \rightarrow H_1(L)$  takes the generator  $\{t^1 + \cdots + t^6\}_{K,L}$  to  $\{e^7 + e^8 + e^9\}_L - \{e^{10} + e^{11} + e^{12}\}_L$ , while  $\partial_* : H_1(K, L) \rightarrow H_0(L)$  takes  $\{e^6\}_{K,L}$  to  $\{v^1\}_L - \{v^2\}_L$ .

(3X) The reader may care to verify that the sequence

$$\begin{aligned} H_2(L) \xrightarrow{i_*} H_2(K) \xrightarrow{j_*} H_2(K, L) \xrightarrow{\partial_*} H_1(L) \xrightarrow{i_*} H_1(K) \\ \xrightarrow{j_*} H_1(K, L) \xrightarrow{\partial_*} H_0(L) \xrightarrow{i_*} H_0(K) \xrightarrow{j_*} H_0(K, L) \rightarrow 0 \end{aligned}$$

is exact in each of the cases (1), (2) above. (Compare 6.1.)

(4X) If  $K$  and  $L$  are both connected, then  $\partial_* : H_1(K, L) \rightarrow H_0(L)$  is always trivial (compare (1) above).

## ★ Appendix on chain complexes

This appendix is optional reading, except that in Chapter 5, on the invariance of homology groups, we make brief use of the first two items, 4.36 and 4.37. The main purpose of the appendix is to sketch a systematic approach to the construction of homomorphisms like the three,  $i_*$ ,  $j_*$  and  $\partial_*$ , constructed above. An elegant generalization of this approach has been given by D. Puppe (1962).

**Definitions 4.36.** A *chain complex*  $(C_p, d_p)$  is a collection  $(C_p)$  of abelian groups, one for each  $p \in \mathbb{Z}$ , together with a homomorphism  $d_p : C_p \rightarrow C_{p-1}$  for each  $p$ , such that, for all  $p$ ,  $d_{p-1} \circ d_p = 0$ . The examples to keep in mind are  $C_p = C_p(K)$ ,  $d_p = \partial_p$  where  $K$  is an oriented simplicial complex, and  $C_p = C_p(K, L)$ ,  $d_p = \hat{\partial}_p$  where  $L$  is a subcomplex of the oriented simplicial complex  $K$ . A *chain map*  $(f_p)$  from a chain complex  $(C_p, d_p)$  to another,  $(C'_p, d'_p)$  is a collection of homomorphisms  $f_p : C_p \rightarrow C'_p$ , such that  $d'_p \circ f_p = f_{p-1} \circ d_p$  for all  $p$ . This means that the diagram

$$\begin{array}{ccccccc} \cdots & \rightarrow & C_{p+1} & \xrightarrow{d_{p+1}} & C_p & \xrightarrow{d_p} & C_{p-1} \rightarrow \cdots \\ & & \downarrow f_{p+1} & & \downarrow f_p & & \downarrow f_{p-1} \\ \cdots & \rightarrow & C'_{p+1} & \xrightarrow{d'_{p+1}} & C'_p & \xrightarrow{d'_p} & C'_{p-1} \rightarrow \cdots \end{array}$$

‘commutes’, i.e. for each square the two routes between the top left and bottom right are in fact the same homomorphism.

We can carry over the terminology of simplicial complexes to the general situation and speak of  $p$ -cycles (elements of  $Z_p = \text{Ker } d_p$ ),  $p$ -boundaries (elements of  $B_p = \text{Im } d_{p+1}$ ) and *homology groups*  $H_p = Z_p/B_p$ , which exist because  $d_{p-1} \circ d_p = 0$ .

Thus a chain map takes cycles to cycles and boundaries to boundaries. It follows easily that there is an induced homomorphism

**4.37**  $f_* : H_p \rightarrow H'_p$   
defined by  $f_*(z + B_p) = f(z) + B'_p \quad (z \in Z_p)$

or, as we may write, using  $\{ \}$  ambiguously,

$$f_*\{z\} = \{f(z)\}.$$

**Examples 4.38.** (1X) Taking, in the usual notation

$$\begin{aligned} C_p &= C_p(L), & d_p &= \partial_p : C_p(L) \rightarrow C_{p-1}(L) \\ C'_p &= C_p(K), & d_p &= \partial_p : C_p(K) \rightarrow C_{p-1}(K) \end{aligned}$$

and  $i_p : C_p(L) \rightarrow C_p(K)$  to be the inclusion,  $(i_p)$  is in fact a chain map, giving an induced homomorphism

$$i_* : H_p(L) \rightarrow H_p(K)$$

as in 4.30.

(2X) Taking

$$\begin{aligned} C_p &= C_p(K), & d_p &= \partial_p : C_p(K) \rightarrow C_{p-1}(K) \\ C'_p &= C_p(K, L), & d_p &= \widehat{\partial}_p : C_p(K, L) \rightarrow C_{p-1}(K, L) \end{aligned}$$

and  $j_p : C_p(K) \rightarrow C_p(K, L)$  to be the usual  $j_p$  (4.20),  $(j_p)$  is in fact a chain map giving an induced homomorphism

$$j_* : H_p(K) \rightarrow H_p(K, L)$$

as in 4.32.

(3X) Show that the ‘restriction’  $\rho$  of the proof of 4.24(2) does not, except in special cases like  $L = K$  or  $L = \emptyset$ , give a chain map between the appropriate chain complexes.

(4XX) Suppose that each  $f_p$  in 4.36 is an epimorphism (surjective). Show that

$$z' + f_p Z_p \rightarrow \{z'\} + \{f_* H_p\} \quad (z' \in Z'_p)$$

defines an isomorphism

$$Z'_p / f_p Z_p \rightarrow H'_p / f_* H_p.$$

(In particular this is true when  $f_p = j_p$ .)

(5X) Let  $f_p : C_p(K, L) \rightarrow C_{p-1}(L)$  be the composite

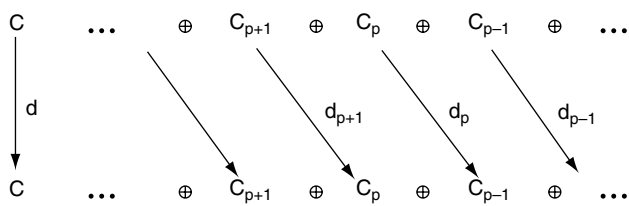
$$C_p(K, L) \xrightarrow{\text{inclusion}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K) \xrightarrow{\rho} C_{p-1}(L).$$

Taking  $C_p = C_p(K, L)$ ,  $d_p = \widehat{\partial}_p$ ;  $C'_p = C_{p-1}(L)$ ,  $d'_p = \partial_{p-1}$  it is not quite true that  $(f_p)$  is a chain map from  $(C_p, d_p)$  to  $(C'_p, d'_p)$ , for in fact  $d'_p \circ f_p = -f_{p-1} \circ d_p$ . Nevertheless  $f_p$  does take cycles to cycles and boundaries to boundaries, and there is an induced homomorphism  $H_p(K, L) \rightarrow H_{p-1}(L)$ , which is in fact the  $\partial_*$  of 4.34.

The notation for chain complexes and chain maps is easier to handle when the chain groups  $C_p$  are collected together into one big group  $C$ , as follows.

**Definitions 4.39.** Let  $(C_p, d_p)$  be a chain complex. The *direct sum*  $C = \oplus C_p$  of the groups  $C_p$  is the abelian group whose elements are families  $(c_p)$ ,  $c_p \in C_p$ ,  $p \in \mathbb{Z}$ , with all but a finite number of  $c_p$  equal to 0, and the operation  $(c_p^1) + (c_p^2) = (c_p^1 + c_p^2)$ . Thus if  $C_p = 0$  when  $p < 0$  or  $p > n$ , as is the case in the geometrical examples, then  $C \cong C_0 \oplus C_1 \oplus \dots \oplus C_n$  and an element of  $C$  can be regarded as an  $n$ -tuple  $(c_0, c_1, \dots, c_n)$ .

The collection of homomorphisms  $(d_p)$  can be brought together into a single homomorphism  $d : C \rightarrow C$ , defined by: the element in the  $p$ th place of the family  $d(c_p)$  is  $d_{p+1}c_{p+1} \in C_p$ .



Then the condition  $d_{p-1} \circ d_p = 0$  for all  $p$  is equivalent to  $d \circ d = 0$ , i.e.  $\text{Im } d \subset \text{Ker } d$ . Also a chain map  $(f_p)$  can be thought of as a single homomorphism  $f : C \rightarrow C'$  such that  $d' \circ f = f \circ d$ .

The group  $C$ , together with  $d : C \rightarrow C$  is sometimes called a *differential graded group*,  $d$  being the differential and the decomposition of  $C$  as a direct sum being the grading. For further information see for example the books of Hilton & Wylie or Prasolov listed in the References. ★